

The `mousik` package*

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Abstract

This is a music-related package focused on writing music fragments. It works specially well with small examples, such as excerpts for theoretical articles. It is easier than other music writing methods in $\text{\LaTeX}$ because it is based on the `Tikz` package.

Keywords

mousik, music, music theory, music notation, music writing, score, sheet, tikz diagram, bar, clef, time signature, note, symbol, rest, dot, chord, accidental, sharp, flat, natural, key

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*This document corresponds to `mousik` v0.1, dated 2020/07/01.

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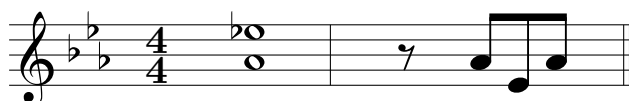
1 Introduction

There are already several ways to set music using Tex. As stated by the ABC package, “*notably MusicTEX and Lilypond*”, as well as ABC per se. Some of them are difficult to use for simple or short fragments; others do not work at all.

The `mousik` package is perfect for short, one line music fragments. It may be used for longer scores, but there are more adaptable packages than this one. However, in spite of its compactness, the package is quite complete and versatile, but easy to learn and use.

2 The `mousik` environment

The way to draw a score is by means of the `mousik` environment.



```

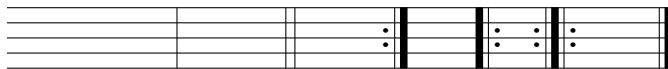
\begin{mousik}
  \Clef{2}
  \Key{-3}
  \TimeSig{4}{4}

  \Note{1}{5}
  \Chord
  \Acc{-1}{9}
  \Note{1}{9}
  \Bar

  \Rest{8}
  \PolyNote{8}{5,2,5}
  \FinalBar
\end{mousik}

```

2.1 Ending a line: bars

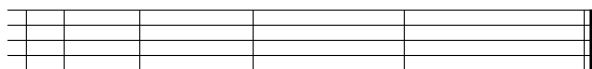


<code>\Space</code>	<code>\Bar</code>	<code>\DoubleBar</code>	<code>\RepeatBar</code>	<code>\FinalBar</code>
		$\{L\}$	$\{R\}$	$\{LR\}$

The only way to end a line in the score is with a bar. Any other way may cause visual errors (see subsection 4.1).

2.2 The \Space command

The `\Space` command is not only used when no bar is needed. It may also be used to input *extra* space between symbols. The default space is 2 (which is the default space between symbols in general), but it may be any whole number (see subsection 4.2).


$$\begin{array}{cccc} \backslash \text{Space}[-1] & \backslash \text{Space}[1] & \backslash \text{Space} & \backslash \text{Space}[3] \\ & \backslash \text{Space}[0] & & \end{array}$$

3 Common symbols

3.1 The `\Clef` command

Clefs might be used at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1).

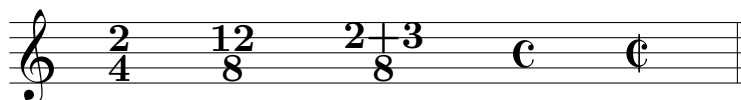
- `\Clef{1}` and `\Clef{2}` are Treble Clefs in the first and second lines.
- `\Clef{3}` and `\Clef{4}` are Bass Clefs in the third and fourth lines.
- `\Clef{11}`, `\Clef{12}`, `\Clef{13}`, `\Clef{14}` and `\Clef{15}` are Alto Clefs in each of the five lines.



3.2 The `\TimeS` command

Time signatures might be used at any moment on the score. `\TimeSig{2}{4}` will draw a 2 by 4 time signature. Any number or symbol can be written inside the brackets.

Also `\TimeSig[1]{}{}` will produce a *common time* symbol and `\TimeSig[2]{}{}` will produce an *alla breve* symbol.



3.3 Notes and rests

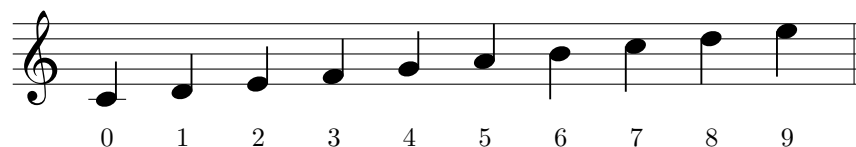
3.3.1 The `\Note` command

The `\Note` command prints a note on the score. It accepts 2 arguments.

The first argument is the duration of the note, from one of these: 1 (whole note / semibreve), 2 (half note / minim), 4 (quarter note / crotchet), 8 (eighth note / quaver), 16 (sixteenth note / semiquaver), 32 (thirty second note / demisemi-quaver).



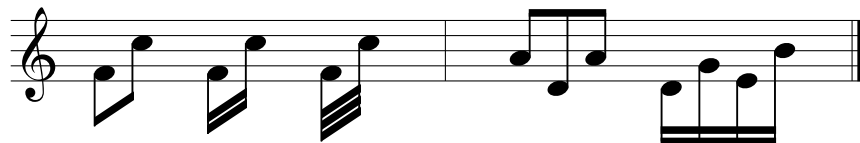
The second argument is the absolute height. For example, these are the notes from 0 to 9:



3.3.2 The `\PolyNote` command

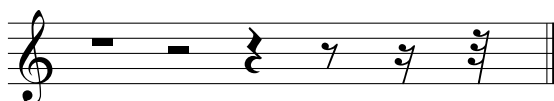
The `\PolyNote` command prints several notes tied together by a beamer. It accepts 2 arguments: the duration of the note and the list of heights separated by commas: `\PolyNote{8}{6,3,6}`.

When there are 2 notes, the beamer is diagonal; when there are more notes, the beamer becomes horizontal.



3.3.3 The `\Rest` command

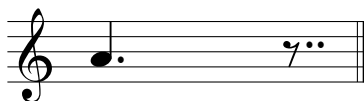
The `\Rest` command prints a rest. It accepts its duration as an argument: 1, 2, 4, 8, 16 or 32.



3.3.4 The `\Dot` command

The `\Dot` command prints any amount of dots in front of a note or rest. It accepts an optional argument and a compulsory argument.

The default number of dots is 1, but it may be any whole number bigger than one. The second, compulsory argument is the height of the dot: `\Dot[2]{7}`.



<code>\Dot{5}</code>	<code>\Dot[2]{7}</code>
<code>\Note{4}{5}</code>	<code>\Rest{8}</code>

3.3.5 The `\Chord` command

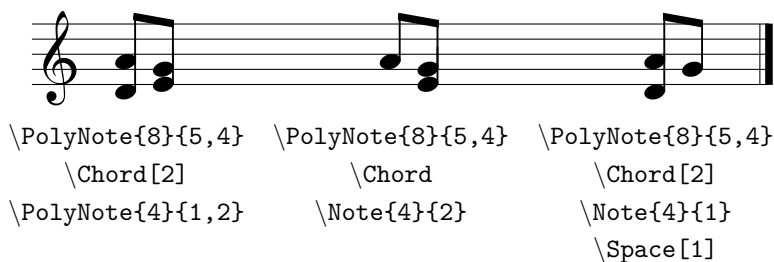
Chords may be obtained by means of the `\Chord` command. It accepts an optional argument: the number of notes to step backwards —mostly for the `\PolyNote` command— with 1 as default.

The best way to use it is to call the `\Note` or `\PolyNote` commands with the highest notes; then call the `\Chord` command; then call the `\Note` or `\PolyNote` commands with the duration 4 of a quarter note, to print them on top.

If the stem needs to be longer, the user may need the `\Enlarge` command, which accepts as arguments the 2 heights of both notes.



For extreme uses of the command, extra `\Space` commands might be needed, as seen here:

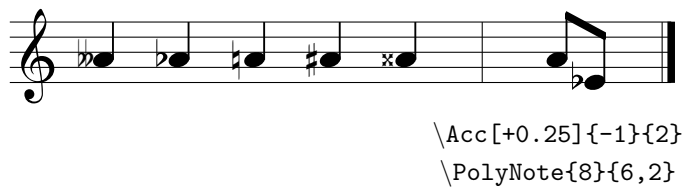


3.4 The `\Acc` command

The `Acc` command prints an accidental before any note. It accepts 2 arguments. The first argument is the type of accidental:

- -2 for the $\flat\flat$ double flat symbol.
- -1 for the $\flat$ flat symbol.
- 0 for the $\natural$ natural symbol.
- 1 for the $\sharp$ sharp symbol.
- 2 for the $\sharp\sharp$ double sharp symbol.

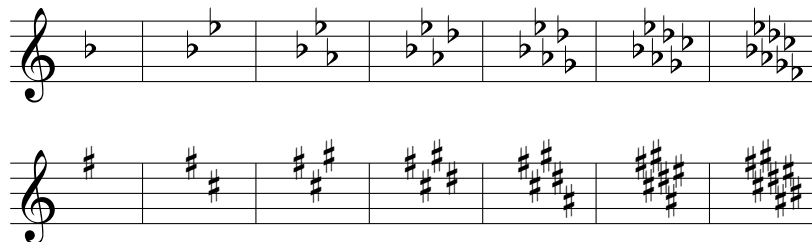
The second argument is the height of the symbol. For usage related to the `\PolyNote` command, an optional argument might be used before the other two, but it is still experimental.



3.5 The `\Key` command

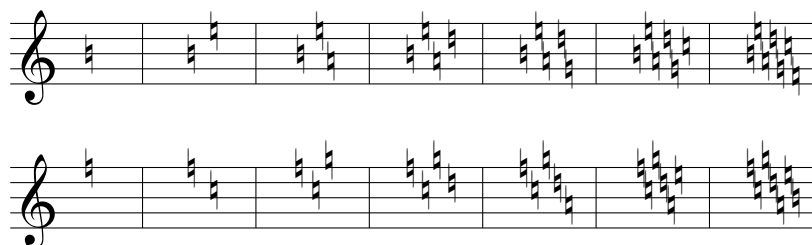
The `\Key` command prints the key at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1).

It accepts an argument: a positive whole number from 1 to 7 for sharps $\sharp$, and a negative whole number from -1 to -7 for flats $\flat$.



3.6 The `\NoKey` command

The `\NoKey` command is the same as the `\Key` command, but it prints naturals $\natural$ instead of sharps or flats.



Be careful: the `\Key` and `\NoKey` commands are affected by the current clef! Nonetheless, other commands such as `\Note` or `\Acc` are not affected and need absolute height values.



```
\begin{mousik}
  \Clef{4}
  \Key{1}
  \TimeSig[2]{}{}

  \Acc{0}{8}
  \Note{1}{8}
  \FinalBar
\end{mousik}
```

3.7 The `\Text` command

The `\Text` command allows the printing of any text string, including dynamics and lyrics. It has a compulsory argument, but also an optional one where the user may choose the height of the string: `\Text[2]{\it rit.}`.



4 Observations

4.1 Overflow of lines



4.2 How do spaces work?

Spaces serve as the manual progression of an invisible cursor during the printing. A negative

5 The package code

```

1  \RequirePackage[minimal]{leadsheets}
   \useleadsheetslibraries {musicsymbols}
3  \RequirePackage{harmony}

5  \RequirePackage{tikz}

7  %%%%%%%%%%%%%%

9  \newcounter{Mevaluate}

11 \newcounter{Mmodulo}
   \newcommand{\MModulo}[2]{
13   \setcounter{Mmodulo}{#1}
   \loop
15   \setcounter{Mevaluate}{#2}\addtocounter{Mevaluate}{-1}
   \ifnum\theMmodulo>\theMevaluate
17     \addtocounter{Mmodulo}{-#2}
     \repeat
19   \ifnum\theMmodulo<0
     \addtocounter{Mmodulo}{#2}
21   \repeat
   }

23 \newcounter{Mmax}
   \newcommand{\MMax}[2][\theMmax]{
25   \setcounter{Mmax}{#1}
27   \ifnum#2>#1
     \setcounter{Mmax}{#2}
29   \fi
   }

31 \newcounter{Mmin}
   \newcommand{\MMin}[2][\theMmin]{
33   \setcounter{Mmin}{#1}
35   \ifnum#2<#1
     \setcounter{Mmin}{#2}
37   \fi
   }

39 %%%%%%%%%%%%%%

41 \newcounter{Mfirst}
43 \newcounter{Mxpos}
   \newcounter{Mypos}
45 \newcounter{Mclef}
   \newcounter{Mkey}

47 \newenvironment{mousik}
49 { \setcounter{Mxpos}{0}
   \setcounter{Mypos}{0}
51   \setcounter{Mclef}{0}
   \setcounter{Mkey}{0}
53   \noindent\begin{tikzpicture}}
   { \addtocounter{Mxpos}{-2}\MStaff
55   \setcounter{Mxpos}{0}
   \setcounter{Mypos}{0}
57   \setcounter{Mclef}{0}

```

```

\setcounter{Mkey}{0}
59 \end{tikzpicture}}

61 \newcommand{\MStaff}{
  \foreach\i in {0,...,\theMypos} {
63    \foreach\j in {0,...,4} {
      \ifnum\i=\theMypos
65        \draw (-0.25,-\i*2+\j*0.2) -- (\theMxpos*0.5,-\i*2+\j*0.2);
      \else
67        \draw (-0.25,-\i*2+\j*0.2) -- (11,-\i*2+\j*0.2);
      \fi
69    }
  }
71 }

73 \newcommand{\Clef}[1]{
  \setcounter{Mclef}{#1}
75 \Adv

77 \ifnum#1=1
  \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.15)
79 {\trebleclef};
\fi\ifnum#1=2
81 \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.35)
{\trebleclef};
83 \fi\ifnum#1=3
  \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.8)
85 {{\resizebox{0.7cm}{0.8cm}{\bassclef}}};
\fi\ifnum#1=4
87 \node at (\theMxpos*0.5+0.1,-\theMypos*2+1)
{{\resizebox{0.7cm}{0.8cm}{\bassclef}}};
89 \fi\ifnum#1=11
  \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.4)
91 {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
\fi\ifnum#1=12
93 \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.6)
{{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
95 \fi\ifnum#1=13
  \node at (\theMxpos*0.5+0.1,-\theMypos*2+0.8)
97 {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
\fi\ifnum#1=14
99 \node at (\theMxpos*0.5+0.1,-\theMypos*2+1)
{{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
101 \fi\ifnum#1=15
  \node at (\theMxpos*0.5+0.1,-\theMypos*2+1.2)
103 {{\resizebox{0.65cm}{0.8cm}{\altoclef}}};
\fi
105 \addtocounter{Mxpos}{2}
107 }

\newlength{\MWidth}
109 \newlength{\MHeight}
\newcommand{\MResize}[1]{
111 \settowidth{\MWidth}{#1}
  \settoheight{\MHeight}{#1}
113 \resizebox{1.5\MWidth}{1.5\MHeight}{#1}
}

```

```

115 \newcommand{\TimeSig}[3][0]{
    \Adv
117     \ifnum#1=0
        \node at (\theMxpos*0.5,-\theMypos*2+0.6) {\bf \MResize{#2}};
119     \node at (\theMxpos*0.5,-\theMypos*2+0.2) {\bf \MResize{#3}};
        \fi \ifnum#1=1
121     \node at (\theMxpos*0.5,-\theMypos*2+0.5) {\meterC};
        \fi \ifnum#1=2
123     \node at (\theMxpos*0.5,-\theMypos*2+0.575) {\allabreve};
        \fi
125     \addtocounter{Mxpos}{3}
    }
127
    \newcommand{\Dot}[2][1]{
129     \foreach \d in {1,...,#1}
        {\node at (\theMxpos*0.5+0.1+0.15*\d,-\theMypos*2-0.21+0.1*#2)
131         {\resizebox{0.1cm}{0.1cm}{\bullet$}};}}
    }
133
    %%%%%%%%%%%
135
    \newcommand{\Head}[1]{
137     \node [opacity=0.0] (tmp) at (\theMxpos*0.5,-\theMypos*2-0.2+#1*0.1) {};
        \fill [rotate around={16:(tmp.center)}] (\theMxpos*0.5-0.01,-\theMypos*2-0.195+#1*0.1) ellipse (4pt and 2.75pt);}
139
    \newcommand{\HollowHead}[1]{
141     \node [opacity=0.0] (tmp) at (\theMxpos*0.5,-\theMypos*2-0.2+#1*0.1) {};
        \fill [black,rotate around={16:(tmp.center)}] (\theMxpos*0.5-0.01,-\theMypos*2-0.195+#1*0.1) ellipse (4pt and 2.75pt);
143     \fill [white,rotate around={19:(tmp.center)}] (\theMxpos*0.5-0.01,-\theMypos*2-0.195+#1*0.1) ellipse (3.2pt and 1.35pt);
    }
145

147 \newcommand{\Enlarge}[3][0]{
    \draw[line width=0.25mm] (\theMxpos*0.5+0.115,-\theMypos*2+#2*0.1-0.2) --
149     (\theMxpos*0.5+0.115,-\theMypos*2+#3*0.1+#1);
    }
151
    \newcommand{\EnlargeDown}[3][0]{
153     \draw[line width=0.25mm] (\theMxpos*0.5-0.14,-\theMypos*2+#2*0.1-0.2) --
        (\theMxpos*0.5-0.14,-\theMypos*2+#3*0.1-0.5-#1);
155 }

157 \newcommand{\Ledger}[1]{
    \ifnum1>#1
159     \foreach\i in {0,...,#1} {
        \ifthenelse{\isodd{\i}}{}{
161         {\draw (\theMxpos*0.5-0.25,-\theMypos*2-0.2+\i*0.1) -- (\theMxpos*0.5+0.25,-\theMypos*2-0.2+\i*0.1);}
        }
163     \fi \ifnum1<#1
        \foreach\i in {12,...,#1} {
165         \ifthenelse{\isodd{\i}}{}{
            {\draw (\theMxpos*0.5-0.25,-\theMypos*2-0.2+\i*0.1) -- (\theMxpos*0.5+0.25,-\theMypos*2-0.2+\i*0.1);}
167         }
        \fi
169 }

171 \newcommand{\Beam}[4]{

```

```

\draw[line width=1mm] (#1*0.5+0.103,-\theMypos*2+#2*0.1#4) -- (\theMxpos*0.5-0.372,-\theMypos*2+#3*0.1#4);
173
\newcommand{\BeamDown}[4]{
175   \draw[line width=1mm] (#1*0.5-0.153,-\theMypos*2+#2*0.1-0.5-#4) -- (\theMxpos*0.5-0.625,-\theMypos*2+#3*0.1#4);
177   \newcommand{\BeamS}[1]{
178     \ifnum\theMfirst=\theMmin
179     \begin{scope}
180       \clip (\theMxpos*0.5-0.4,-\theMypos*2+\theMmin*0.1-0.3) rectangle (\theMxpos*0.5+0.125,-\theMypos*2+0.05+\theMmin*0.1-0.3);
181       \draw[line width=1mm] (\theMxpos*0.5-0.411,-\theMypos*2+\theMmin*0.1#1) -- (\theMxpos*0.5+0.147,-\theMypos*2+0.05+\theMmin*0.1-0.3);
182     \end{scope}
183     \else
184     \begin{scope}
185       \clip (\theMxpos*0.5-0.4,-\theMypos*2+0.05+\theMmax*0.1#1) rectangle (\theMxpos*0.5+0.125,-\theMypos*2+0.05+\theMmax*0.1-0.3);
186       \draw[line width=1mm] (\theMxpos*0.5-0.411,-\theMypos*2+\theMmax*0.1#1) -- (\theMxpos*0.5+0.147,-\theMypos*2+0.05+\theMmax*0.1-0.3);
187     \end{scope}
188     \fi
189   }

191   \newcommand{\BeamSDown}[1]{
192     \ifnum\theMfirst=\theMmin
193     \begin{scope}
194       \clip (\theMxpos*0.5-0.645,-\theMypos*2+\theMmin*0.1-0.5-#1) rectangle (\theMxpos*0.5-0.129,-\theMypos*2+\theMmin*0.1-0.5-#1);
195       \draw[line width=1mm] (\theMxpos*0.5-0.7,-\theMypos*2+\theMmin*0.1-#1-0.48) -- (\theMxpos*0.5-0.1,-\theMypos*2+\theMmin*0.1-0.5-#1);
196     \end{scope}
197     \else
198     \begin{scope}
199       \clip (\theMxpos*0.5-0.65,-\theMypos*2+\theMmin*0.1-0.5-#1) rectangle (\theMxpos*0.5-0.129,-\theMypos*2+\theMmin*0.1-0.5-#1);
200       \draw[line width=1mm] (\theMxpos*0.5-0.7,-\theMypos*2+\theMmax*0.1-#1-0.48) -- (\theMxpos*0.5-0.1,-\theMypos*2+\theMmax*0.1-0.5-#1);
201     \end{scope}
202     \fi
203   }

205   \newcommand{\Semibreve}[1]{
206     \node at (\theMxpos*0.5,-\theMypos*2-0.2+#1*0.1) {\resizebox{0.3cm}{0.2cm}{\Ganz}};
207   }
208   \newcommand{\Minim}[1]{
209     \HollowHead{#1}
210     \ifnum#1<6
211       \Enlarge[0.3]{#1}{#1}
212     \else
213       \EnlargeDown[0.3]{#1}{#1}
214     \fi
215   }
216   \newcommand{\Crotchet}[1]{
217     \Head{#1}
218     \ifnum#1<6
219       \Enlarge[0.3]{#1}{#1}
220     \else
221       \EnlargeDown[0.3]{#1}{#1}
222     \fi
223   }% TODO
224   \newcommand{\Quaver}[1]{
225     \Head{#1}
226     \ifnum#1<6
227       \Enlarge[0.4]{#1}{#1}
228     \else

```

```

229     \EnlargeDown[0.4]{#1}{#1}
231     \fi
232 }
233 \newcommand{\Semiquaver}[1]{
234     \Head{#1}
235     \ifnum#1<6
236         \Enlarge[0.5]{#1}{#1}
237     \else
238         \EnlargeDown[0.5]{#1}{#1}
239     \fi
240 }
241 \newcommand{\Demisemiquaver}[1]{
242     \Head{#1}
243     \ifnum#1<6
244         \Enlarge[0.6]{#1}{#1}
245     \else
246         \EnlargeDown[0.6]{#1}{#1}
247     \fi
248 }
249 \newcommand{\Note}[2]{
250     \Adv
251     \ifnum#1=1
252         \Semibreve{#2}
253     \fi \ifnum#1=2
254         \Minim{#2}
255     \fi \ifnum#1=4
256         \Crotchet{#2}
257     \fi \ifnum#1=8 %%% TODO
258         \node at (\theMxpos*0.5+0.09,-\theMypos*2+0.02+#2*0.1) {\resizebox{0.45cm}{0.65cm}{\Acht}};
259     \fi \ifnum#1=16
260         \node at (\theMxpos*0.5+0.09,-\theMypos*2+0.02+#2*0.1) {\resizebox{0.45cm}{0.65cm}{\Sech}};
261     \fi \ifnum#1=32
262         \node at (\theMxpos*0.5+0.09,-\theMypos*2+0.095+#2*0.1) {\resizebox{0.45cm}{0.8cm}{\Zwdr}};
263     \fi
264     \Ledge{#2}
265     \addtocounter{Mxpos}{2}
266 }
267 }
268
269 \newcommand{\PolyNote}[2]{
270     \Adv
271     \setcounter{Mfirst}{-50}
272     \setcounter{Mmax}{-50}
273     \setcounter{Mmin}{50}
274     \setcounter{aux}{\theMxpos}
275
276     \ifnum#1=1
277         \foreach\ n in {#2} {
278             \Semibreve{\ n}
279             \stepcounter{Mxpos}
280         }
281     \else \ifnum#1=2
282         \foreach\ n in {#2} {
283             \node at (\theMxpos*0.5,-\theMypos*2-0.2+\ n*0.1) {\resizebox{0.3cm}{0.2cm}{\Halb}};
284             \stepcounter{Mxpos}
285         }
286     \else \ifnum#1=4

```

```

287     \foreach\n in {#2} {
        \Crotchet{\n}
        \stepcounter{Mxpos}}
289     \else
        \foreach\n in {#2} {
291             \Head{\n}
                \stepcounter{Mxpos}
293             \MMax{\n}\MMin{\n}
                \ifnum\theMfirst=-50\setcounter{Mfirst}{\n}\fi
295         }
        \setcounter{Mevaluate}{\theMxpos}\addtocounter{Mevaluate}{-\theaux}
297     \setcounter{Mxpos}{\theaux}
        \ifnum\theMevaluate=2
299         \foreach\n in {#2} {
            \ifnum\theMmax<6
301                 \ifnum#1=8
                    \Enlarge[0.4]{\n}{\n}
303                 \fi \ifnum#1=16
                    \Enlarge[0.5]{\n}{\n}
305                 \fi \ifnum#1=32
                    \Enlarge[0.6]{\n}{\n}
307                 \fi
            \else
309                 \ifnum#1=8
                    \EnlargeDown[0.4]{\n}{\n}
311                 \fi \ifnum#1=16
                    \EnlargeDown[0.5]{\n}{\n}
313                 \fi \ifnum#1=32
                    \EnlargeDown[0.6]{\n}{\n}
315                 \fi
            \fi
317         \stepcounter{Mxpos}
        }
319     \addtocounter{Mxpos}{-1}

321     \ifnum\theMmax<6
        \ifnum#1=8
323             \BeamS{+0.4}
        \fi \ifnum#1=16
325             \BeamS{+0.5}
            \BeamS{+0.35}
327        \fi \ifnum#1=32
            \BeamS{+0.6}
329            \BeamS{+0.45}
            \BeamS{+0.3}
331        \fi
    \else
333        \ifnum#1=8
            \BeamSDown{+0.4}
335        \fi \ifnum#1=16
            \BeamSDown{+0.5}
337            \BeamSDown{+0.35}
        \fi \ifnum#1=32
339            \BeamSDown{+0.6}
            \BeamSDown{+0.45}
341            \BeamSDown{+0.3}
        \fi

```

```

343 \fi
\else
345 \foreach\n in {\#2} {
\ifnum\theMmax<6
347 \ifnum#1=8
\Enlarge[0.4]{\n}{\theMmax}
349 \fi\ifnum#1=16
\Enlarge[0.5]{\n}{\theMmax}
351 \fi\ifnum#1=32
\Enlarge[0.6]{\n}{\theMmax}
353 \fi
\else
355 \ifnum#1=8
\EnlargeDown[0.3]{\n}{\theMmin}
357 \fi\ifnum#1=16
\EnlargeDown[0.4]{\n}{\theMmin}
359 \fi\ifnum#1=32
\EnlargeDown[0.5]{\n}{\theMmin}
361 \fi
\fi
363 \stepcounter{Mxpos}
}
365 \ifnum\theMmax<6
\ifnum#1=8
367 \Beam{\theaux}{\theMmax}{\theMmax}{+0.4}
\fi\ifnum#1=16
369 \Beam{\theaux}{\theMmax}{\theMmax}{+0.5}
\Beam{\theaux}{\theMmax}{\theMmax}{+0.35}
371 \fi\ifnum#1=32
\Beam{\theaux}{\theMmax}{\theMmax}{+0.6}
373 \Beam{\theaux}{\theMmax}{\theMmax}{+0.45}
\Beam{\theaux}{\theMmax}{\theMmax}{+0.3}
375 \fi
\else
377 \ifnum#1=8
\BeamDown{\theaux}{\theMmin}{\theMmin}{+0.3}
379 \fi\ifnum#1=16
\BeamDown{\theaux}{\theMmin}{\theMmin}{+0.4}
381 \BeamDown{\theaux}{\theMmin}{\theMmin}{+0.25}
\fi\ifnum#1=32
383 \BeamDown{\theaux}{\theMmin}{\theMmin}{+0.5}
\BeamDown{\theaux}{\theMmin}{\theMmin}{+0.35}
385 \BeamDown{\theaux}{\theMmin}{\theMmin}{+0.2}
\fi
387 \fi
\fi
389 \fi\fi\fi
\setcounter{Mxpos}{\theaux}
391 \foreach\n in {\#2} {
\Ledger{\n}
393 \stepcounter{Mxpos}
}
395 \addtocounter{Mxpos}{1}
}
397
\newcommand{\Rest}[1]{
399 \Adv

```

```

401 \ifnum#1=1
\node at (\theMxpos*0.5,-\theMypos*2+0.52) {\resizebox{0.3cm}{0.15cm}{\GaPa}};
403 \fi\ifnum#1=2
\node at (\theMxpos*0.5,-\theMypos*2+0.42) {\resizebox{0.3cm}{0.15cm}{\HaPa}};
405 \fi\ifnum#1=4
\node at (\theMxpos*0.5,-\theMypos*2+0.4) {\resizebox{0.3cm}{0.7cm}{\ViPa}};
407 \fi\ifnum#1=8
\node at (\theMxpos*0.5,-\theMypos*2+0.32) {\resizebox{0.3cm}{0.45cm}{\AcPa}};
409 \fi\ifnum#1=16
\node at (\theMxpos*0.5,-\theMypos*2+0.3) {\resizebox{0.35cm}{0.525cm}{\SePa}};
411 \fi\ifnum#1=32
\node at (\theMxpos*0.5,-\theMypos*2+0.375) {\resizebox{0.4cm}{0.6cm}{\ZwPa}};
413 \fi

415 \addtocounter{Mxpos}{2}
}

417 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

419 \newcommand{\Acc}[3][−0.25]{
421 \ifnum#2=0
\node at (\theMxpos*0.5#1,-\theMypos*2−0.2+0.1*#3)
423 {{\resizebox{0.15cm}{0.35cm}{\natural}}};
\fi\ifnum#2=1
425 \node at (\theMxpos*0.5#1,-\theMypos*2−0.2+0.1*#3)
{{\resizebox{0.15cm}{0.35cm}{\sharp}}};
427 \fi\ifnum#2=−1
\node at (\theMxpos*0.5#1,-\theMypos*2−0.1+0.1*#3)
429 {{\resizebox{0.15cm}{0.35cm}{\flat}}};
\fi\ifnum#2=2
431 \node at (\theMxpos*0.5#1,-\theMypos*2−0.205+0.1*#3)
{{\resizebox{0.15cm}{0.15cm}{\doublesharp}}};
433 \fi\ifnum#2=−2
\node at (\theMxpos*0.5#1,-\theMypos*2−0.1+0.1*#3)
435 {{\resizebox{0.2cm}{0.35cm}{\doubleflat}}};
\fi
437 }

439 \newcounter{aux}
\newcommand{\AccFN}[2][−0.25]{
441 \KB{#2}
\Acc[#1]{0}{\theaux}
443 \newcommand{\AccSN}[2][−0.25]{
\KS{#2}
445 \Acc[#1]{0}{\theaux}
\newcommand{\AccF}[2][−0.25]{
447 \KB{#2}
\Acc[#1]{−1}{\theaux}
449 \newcommand{\AccS}[2][−0.25]{
\KS{#2}
451 \Acc[#1]{1}{\theaux}

453 \newcommand{\KB}[1]{
\setcounter{aux}{#1}
455 \addtocounter{aux}{−3}

```

```

457 \ifnum\theMclef=1
    \addtocounter{aux}{-2}
459 \fi\ifnum\theMclef=2
    \addtocounter{aux}{0}
461 \fi\ifnum\theMclef=3
    \addtocounter{aux}{3}
463 \fi\ifnum\theMclef=4
    \addtocounter{aux}{-2}
465 \fi\ifnum\theMclef=11
    \addtocounter{aux}{2}
467 \fi\ifnum\theMclef=12
    \addtocounter{aux}{-3}
469 \fi\ifnum\theMclef=13
    \addtocounter{aux}{-1}
471 \fi\ifnum\theMclef=14
    \addtocounter{aux}{1}
473 \fi\ifnum\theMclef=15
    \addtocounter{aux}{3}
475 \fi
    \MModulo{\theaux}{7}
477 \setcounter{aux}{\theMmodulo}
    \addtocounter{aux}{3}
479 }
    \newcommand{\KS}[1]{
481 \setcounter{aux}{#1}
    \addtocounter{aux}{-5}
483
    \ifnum\theMclef=1
485 \addtocounter{aux}{-2}
    \fi\ifnum\theMclef=2
487 \addtocounter{aux}{0}
    \fi\ifnum\theMclef=3
489 \addtocounter{aux}{3}
    \fi\ifnum\theMclef=4
491 \addtocounter{aux}{-2}
    \fi\ifnum\theMclef=11
493 \addtocounter{aux}{2}
    \fi\ifnum\theMclef=12
495 \addtocounter{aux}{-3}
    \fi\ifnum\theMclef=13
497 \addtocounter{aux}{-1}
    \fi\ifnum\theMclef=14
499 \addtocounter{aux}{1}
    \fi\ifnum\theMclef=15
501 \addtocounter{aux}{3}
    \fi
503 \MModulo{\theaux}{7}
    \setcounter{aux}{\theMmodulo}
505 \addtocounter{aux}{5}
    }
507
    \newcommand{\Key}[1]{
509 \setcounter{Mkey}{#1}
    \ifnum#1=-1
511 \AccF[-0.2]{6}
    \fi\ifnum#1=-2
513 \AccF[-0.35]{6}

```

```

515      \AccF[-0.05]{9}
      \fi \ifnum#1=-3
517      \AccF[-0.4]{6}
      \AccF[-0.15]{9}
      \AccF[-0.0]{5}
519      \fi \ifnum#1=-4
      \AccF[-0.45]{6}
521      \AccF[-0.25]{9}
      \AccF[-0.1]{5}
523      \AccF[+0.1]{8}
      \fi \ifnum#1=-5
525      \AccF[-0.45]{6}
      \AccF[-0.25]{9}
527      \AccF[-0.15]{5}
      \AccF[+0.05]{8}
529      \AccF[+0.15]{4}
      \fi \ifnum#1=-6
531      \AccF[-0.45]{6}
      \AccF[-0.3]{9}
533      \AccF[-0.2]{5}
      \AccF[-0.05]{8}
535      \AccF[+0.05]{4}
      \AccF[+0.2]{7}
537      \fi \ifnum#1=-7
      \AccF[-0.45]{6}
539      \AccF[-0.35]{9}
      \AccF[-0.25]{5}
541      \AccF[-0.15]{8}
      \AccF[-0.05]{4}
543      \AccF[+0.05]{7}
      \AccF[+0.15]{3}
545      \fi
      \ifnum#1=1
547      \AccS[-0.2]{10}
      \fi \ifnum#1=2
549      \AccS[-0.35]{10}
      \AccS[-0.05]{7}
551      \fi \ifnum#1=3
      \AccS[-0.425]{10}
553      \AccS[-0.2]{7}
      \AccS[-0.025]{11}
555      \fi \ifnum#1=4
      \AccS[-0.45]{10}
557      \AccS[-0.25]{7}
      \AccS[-0.1]{11}
559      \AccS[+0.1]{8}
      \fi \ifnum#1=5
561      \AccS[-0.45]{10}
      \AccS[-0.3]{7}
563      \AccS[-0.15]{11}
      \AccS[-0.0]{8}
565      \AccS[+0.15]{5}
      \fi \ifnum#1=6
567      \AccS[-0.4]{10}
      \AccS[-0.3]{7}
569      \AccS[-0.2]{11}
      \AccS[-0.1]{8}

```

```

571 \AccS[+0.0]{5}
    \AccS[+0.1]{9}
573 \fi \ifnum#1=7
    \AccS[-0.45]{10}
575 \AccS[-0.35]{7}
    \AccS[-0.25]{11}
577 \AccS[-0.15]{8}
    \AccS[-0.05]{5}
579 \AccS[+0.05]{9}
    \AccS[+0.15]{6}
581 \fi
    \stepcounter{Mxpos}
583 }

585 \newcommand{\NoKey}[1]{
    \setcounter{Mkey}{0}
587 \ifnum#1=-1
    \AccFN[-0.2]{6}
589 \fi \ifnum#1=-2
    \AccFN[-0.35]{6}
591 \AccFN[-0.05]{9}
    \fi \ifnum#1=-3
593 \AccFN[-0.4]{6}
    \AccFN[-0.15]{9}
595 \AccFN[-0.0]{5}
    \fi \ifnum#1=-4
597 \AccFN[-0.45]{6}
    \AccFN[-0.25]{9}
599 \AccFN[-0.1]{5}
    \AccFN[+0.1]{8}
601 \fi \ifnum#1=-5
    \AccFN[-0.45]{6}
603 \AccFN[-0.25]{9}
    \AccFN[-0.15]{5}
605 \AccFN[+0.05]{8}
    \AccFN[+0.15]{4}
607 \fi \ifnum#1=-6
    \AccFN[-0.45]{6}
609 \AccFN[-0.3]{9}
    \AccFN[-0.2]{5}
611 \AccFN[-0.05]{8}
    \AccFN[+0.05]{4}
613 \AccFN[+0.2]{7}
    \fi \ifnum#1=-7
615 \AccFN[-0.45]{6}
    \AccFN[-0.35]{9}
617 \AccFN[-0.25]{5}
    \AccFN[-0.15]{8}
619 \AccFN[-0.05]{4}
    \AccFN[+0.05]{7}
621 \AccFN[+0.15]{3}
    \fi
623 \ifnum#1=1
    \AccSN[-0.2]{10}
625 \fi \ifnum#1=2
    \AccSN[-0.35]{10}
627 \AccSN[-0.05]{7}

```

```

\ fi \ ifnum#1=3
629 \AccSN[-0.425]{10}
\AccSN[-0.2]{7}
631 \AccSN[-0.025]{11}
\ fi \ ifnum#1=4
633 \AccSN[-0.45]{10}
\AccSN[-0.25]{7}
635 \AccSN[-0.1]{11}
\AccSN[+0.1]{8}
637 \ fi \ ifnum#1=5
\AccSN[-0.45]{10}
639 \AccSN[-0.3]{7}
\AccSN[-0.15]{11}
641 \AccSN[-0.0]{8}
\AccSN[+0.15]{5}
643 \ fi \ ifnum#1=6
\AccSN[-0.4]{10}
645 \AccSN[-0.3]{7}
\AccSN[-0.2]{11}
647 \AccSN[-0.1]{8}
\AccSN[+0.0]{5}
649 \AccSN[+0.1]{9}
\ fi \ ifnum#1=7
651 \AccSN[-0.45]{10}
\AccSN[-0.35]{7}
653 \AccSN[-0.25]{11}
\AccSN[-0.15]{8}
655 \AccSN[-0.05]{5}
\AccSN[+0.05]{9}
657 \AccSN[+0.15]{6}
\ fi
659 \stepcounter{Mxpos}
}
661 %%%%%%%%%%%
663 \newcommand{\Text}[2][0]{
665 \node at (\theMxpos*0.5,-\theMypos*2-0.7+#1) {\#2};
}
667 \newcommand{\Chord}[1][1]{
669 \addtocounter{Mxpos}{-1}
\addtocounter{Mxpos}{-#1}
671 \newcommand{\Space}[1][2]{\Adv\addtocounter{Mxpos}{#1}}

673 \newcommand{\Bar}{
\ ifnum\theMxpos=23
675 \addtocounter{Mxpos}{-1}
\ fi
677 \draw (\theMxpos*0.5,-\theMypos*2+0.8) -- (\theMxpos*0.5,-\theMypos*2);
\addtocounter{Mxpos}{2}}
679 \newcommand{\DoubleBar}{
681 \ ifnum\theMxpos=23
\addtocounter{Mxpos}{-1}
683 \ fi
\draw (\theMxpos*0.5-0.06,-\theMypos*2+0.8) -- (\theMxpos*0.5-0.06,-\theMypos*2);

```

```

685     \draw (\theMxpos*0.5+0.06,-\theMypos*2+0.8) -- (\theMxpos*0.5+0.06,-\theMypos*2);
        \addtocounter{Mxpos}{2}
687
\newcommand{\FinalBar}{
689     \ifnum\theMxpos=23
        \addtocounter{Mxpos}{-1}
691     \fi
        \draw (\theMxpos*0.5-0.12,-\theMypos*2+0.8) -- (\theMxpos*0.5-0.12,-\theMypos*2);
693     \draw[line width=1mm] (\theMxpos*0.5,-\theMypos*2+0.805) -- (\theMxpos*0.5,-\theMypos*2-0.005);
        \addtocounter{Mxpos}{2}
695 }

697 \newcommand{MString}[3]{%
    \ifthenelse{\equal{#1}{#2}}
699     {#3}
    {}
701 }

703 \newcommand{\RepeatBar}[1]{
    \ifnum\theMxpos=23
705     \addtocounter{Mxpos}{-1}
    \fi
707     MString{#1}{L}{
        \node at (\theMxpos*0.5-0.24,0.4-\theMypos*2) {\resizebox{0.175cm}{0.25cm}{:}};
709         \draw (\theMxpos*0.5-0.12,-\theMypos*2+0.8) -- (\theMxpos*0.5-0.12,-\theMypos*2);
        \draw[line width=1mm] (\theMxpos*0.5,-\theMypos*2+0.805) -- (\theMxpos*0.5,-\theMypos*2-0.005);
711     }
        MString{#1}{R}{
713         \draw[line width=1mm] (\theMxpos*0.5,-\theMypos*2+0.805) -- (\theMxpos*0.5,-\theMypos*2-0.005);
        \node at (\theMxpos*0.5+0.24,0.4-\theMypos*2) {\resizebox{0.175cm}{0.25cm}{:}};
715         \draw (\theMxpos*0.5+0.12,-\theMypos*2+0.8) -- (\theMxpos*0.5+0.12,-\theMypos*2);
        }
717     MString{#1}{LR}{
        \node at (\theMxpos*0.5-0.24,0.4-\theMypos*2) {\resizebox{0.175cm}{0.25cm}{:}};
719         \draw (\theMxpos*0.5-0.12,-\theMypos*2+0.8) -- (\theMxpos*0.5-0.12,-\theMypos*2);
        \draw[line width=1mm] (\theMxpos*0.5,-\theMypos*2+0.805) -- (\theMxpos*0.5,-\theMypos*2-0.005);
721         \node at (\theMxpos*0.5+0.24,0.4-\theMypos*2) {\resizebox{0.175cm}{0.25cm}{:}};
        \draw (\theMxpos*0.5+0.12,-\theMypos*2+0.8) -- (\theMxpos*0.5+0.12,-\theMypos*2);
723     }
        \addtocounter{Mxpos}{2}
725 }

727 \newcommand{\Adv}{
    \ifnum\theMxpos>20
729     \addtocounter{Mypos}{1}
        \setcounter{Mxpos}{0}
731     \ifnum\theMclef=0\else
        \Clef{\theMclef}
733     \fi
        \ifnum\theMkey=0\else
735     \Key{\theMkey}
        \fi
737     \fi
}

```